Phosphine can also prepared by the hydrolysis of metallic phosphides with water or dilute mineral acids.

$$Ca_3P_2 + 6H_2O \longrightarrow \underset{\text{Phosphine}}{2PH_3\uparrow} + 3Ca(OH)_2$$

$$AlP + 3HCl \longrightarrow \underset{\text{Phosphine}}{PH_3\uparrow} + AlCl_3$$

Phosphine is prepared in pure form by heating phosphorous acid.

$$\underset{\text{Phosphorous acid}}{4H_3PO_3} \xrightarrow{\Delta} \underset{\text{Ortho phosphoric acid}}{3H_3PO_4} + \underset{\text{Phosphine}}{PH_3\uparrow}$$

A pure sample of phosphine is prepared by heating phosphonium iodide with caustic soda solution.

$$PH_4I + NaOH \xrightarrow{\Delta} \underset{\text{Phosphine}}{PH_3\uparrow} + NaI + H_2O$$

Physical properties:

It is colourless, poisonous gas with rotten fish smell. It is slightly soluble in water and is neutral to litmus test. It condenses to a colourless liquid at 188 K and freezes to a solid at 139.5 K .

Chemical properties:

Thermal stability: Phosphine decomposes into its elements when heated in absence of air at 317 K or when electric current is passed through it.

$$4PH_3 \xrightarrow{317K} P_4 + 6H_2$$

Combustion: When phosphine is heated with air or oxygen it burns to give meta phosphoric acid.

$$4PH_3 + 8O_2 \xrightarrow{\Delta} \underset{\text{Phosphorous pentoxide}}{P_4O_{10}} + 6H_2O$$

$$P_4O_{10} + 6H_2O \xrightarrow{\Delta} \underset{\text{Meta phosphoric acid}}{4HPO_3} + 4H_2O$$

Basic nature: Phosphine is weakly basic and forms phosphonium salts with halogen acids.

$$PH_3 + HI \longrightarrow PH_4I$$

$$PH_4I + H_2O \xrightarrow{\Delta} \underset{\text{Phosphine}}{PH_3} + H_3O^+ + I^-$$

It reacts with halogens to give phosphorus penta halides.

$$PH_3 + 4Cl_2 \longrightarrow PCl_5 + 3HCl$$

Reducing property : Phosphine precipitates some metal as phosphide from their salt solutions.

$$3AgNO_3 + PH_3 \longrightarrow Ag_3P + 3HNO_3$$

It forms coordination compounds with lewis acids such as boron trichloride.

$$BCl_3 + PH_3 \longrightarrow \underset{\text{Coordination compound}}{[Cl_3B \leftarrow :PH_3]}$$

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